Raif Karaahmetoğlu

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Skills

- MATLAB/Simulink | C | C++ | Python | JavaScript | TypeScript | CSS | HTML | React | React Router | MUI
- Jira/Confluence | AWS | Git | GitHub CI/CD | Scrum/Agile | Enterprise Architect | Docker and Kubernetes (training)
- CATIA | CarMaker | GT-SUITE | AVL Puma | AVL Concerto | AVL Testbed.Connect | AVL Model.CONNECT | AVL fmi.lab
- OOP | SOLID Principles | Design Patterns | Unit Testing

Experience

Software Development Engineer AVL MTC Motortestcenter AB *Stockholm, Sweden* **09/2024 – Present**

- **Scania, Connected and Autonomous Systems** (consultancy assignment, 13 months): Contributed to a Python-based mapping application that generates drivable areas from LiDAR and GPS data, enabling autonomous non-road mining vehicles to navigate safely within mine sites. Redesigned the threaded data download procedure. Introduced the MVVM pattern to separate the UI from the logic in the new map release procedure. Automated the build and installation process through GitHub CI/CD pipelines. Also ensured auxiliary C++ projects comply with MISRA C++:2023.
- **Volvo Construction Equipment** (consultancy assignment, over 3 months): Conducted systems engineering activities for auxiliary functions of haulers, including windshield wiper control and IMU self-calibration.

Application Engineer AVL TR Mühendislik ve Test Sistemleri *İstanbul, Türkiye* **07/2018 – 09/2024**

- Integrated EPA's Greenhouse gas Emissions Model (GEM) into a real power-pack (engine and transmission) test rig to comply with 40 CFR Parts 1036 and 1037. Restructured the model framework, adapted the parameterization process and implemented automation and safety enhancements to the test rig. Introduced on-the-fly updating of model parameters during real-time operation.
- Modernized the ISAC real-time vehicle simulation library by developing contemporary vehicle architecture models. Overcame challenges such as integrating a no-slip model and ensuring stable vehicle control unit (VCU) development. Defined the process for transitioning models from online to offline. Strengthened ISAC's position in the real-time simulation market, contributing to multi-million-euro annual revenue.
- Developed vehicle subsystem models in Python for the Route Studio project, applying the Factory, Builder, Strategy and Facade etc. design patterns to keep the codebase robust and structured.
- Refactored the Model Factory project and integrated it into Route Studio, enabling battery model parameters to be identified from road data and reducing reliance on battery test-cycle data.
- Restructured and enhanced the probabilistic Real Driving Emissions (RDE) cycle generation algorithm and integrated it into Route Studio.
- Contributed to the Functional Mock-up Interface (FMI) module of the VTS project for importing FMUs. Also developed the Model.CONNECT (MC) module, which generates an FMU for real-time monitoring and signal state transfer via WebSockets.
- Developed an automated hardware-in-the-loop (HIL) project creation module for VTS using the COM APIs of dSPACE ConfigurationDesk, ControlDesk and Bus Manager.

Electronic Design Supervisor Hema Endüstri A.Ş. *Tekirdağ, Türkiye* **08/2016 – 02/2018**

- Coordinated negotiations with product development companies on co-developing an electro-hydraulic steering control unit.
- Managed the Test Systems Department, which covered test benches ranging from hydraulic systems to transmission units. Within one year, recommissioned all inoperable benches, 66% of the total, and restored them to full operation. Established guidelines for new test bench development to prevent problems such as incorrect component selection, complex wiring, unmarked cables, signal interference and convoluted algorithm design. Mentored the test engineers.
- Led the Bluetooth-controlled electro-hydraulic lift project, including requirements gathering and documentation. Completed the project and showcased the system at several exhibitions.
- Developed steering system dynamics models for HIL simulators.
- Delivered a prototype automatic guidance system and its control algorithms for agricultural vehicles under a tight deadline, covering latitude/longitude-to-UTM conversion, upper- and lower-level control algorithms and a hydraulic valve controller. Built the application-level algorithms in MATLAB/Simulink and generated C code with Embedded Coder, then wrote the lower-level code in C and integrated both on a TI TMS570LS4032 processor.

R&D Engineer Tümosan A.Ş. *İstanbul, Türkiye* **10/2013 – 01/2015**

- Wrote a project proposal for developing diesel engine control algorithms, which TÜBİTAK TEYDEB accepted and funded with €1.5 million. The budget set up the company's control and electronics laboratory, including the required software, tools and equipment.
- Developed mean-value and zero-dimensional simulation models for diesel engines and established the company's in-house expertise in real-time engine modeling.
- Managed the integration of basic electronic control units into diesel engines to meet Stage IIIB emissions regulations.
- Managed a three-person control software team.

Electromechanics Design Engineer Otokar A.Ş. *Sakarya, Türkiye* **03/2012 – 08/2013**

- Designed the mechanical infrastructure for turrets and remote-controlled weapon stations (RCWS) mounted on ground defense vehicles, including the UCOK compact machine-gun turret, which was added to the company's product portfolio.

Volunteer Researcher Mekar Laboratory *İstanbul, Türkiye* **06/2010 – 08/2011**

- Developed the semi-autonomous vehicle infrastructure, mathematical model and longitudinal control algorithm for the team's entry in the Grand Cooperative Driving Challenge in the Netherlands.

Education

Master of Science İstanbul Technical University *İstanbul, Türkiye* **2009 – 2011**

- Major in Mechatronics Engineering, GPA 3.63/4

Bachelor of Science Eskişehir Osmangazi University *Eskişehir, Türkiye* **2004 – 2009**

- Major in Mechanical Engineering, GPA 3.23/4

Publications

- Cooperative Adaptive Cruise Control Implementation of Team Mekar at the Grand Cooperative Driving Challenge, IEEE Transactions on Intelligent Transportation Systems, vol. 13, no. 3, 2012.
- Design and Control Basics of a Cooperative Vehicle, 6th Automotive Technologies Congress (OTEKON), Bursa, 2012.
- Car Following with Adaptive Cruise Control Evaluated in Istanbul City Traffic Conditions, 7th Automotive Technologies Congress (OTEKON), Bursa, 2014.

Others

- High Honor Student, GPA 3.86 (2007–2008 spring term)